

COMBINATORICS / INTEGER SEQUENCES / RECURRENCES

A sharp order-five threshold for log-concavity of higher-order Stirling subset numbers

An elementary recurrence proof, an exact fifth-order certificate,
and applications to Stirling cycle triangles

Merged research report prepared for Vladimir Reshetnikov

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Abstract

For an integer $r \geq 1$, let $S_{n,k}^{(r)}$ count partitions of an $(n + (r-1)k)$ -element set into k blocks, each of size at least r . We prove that all rows of this diagonal-shifted Stirling triangle are log-concave if and only if $1 \leq r \leq 5$, and that for those orders every internal inequality with $n \geq 2$ and $1 \leq k \leq n-1$ is strict. This establishes the log-concavity assertion of Deb–Sokal’s Conjecture 1.4(c), including its cases $r = 3, 4, 5$, and nothing else of that conjecture. The proof reduces a two-term recurrence to a quadratic form. For $r = 5$ the decisive step is the exact identity $25(A_4u^2 + H_4uv + v^2) = (5v - 8\alpha u)^2 + 96(t-5)\beta\gamma u^2 + 480\beta(t-5k)uv$, whose three summands are nonnegative precisely on the triangular domain; it also supplies an explicit lower bound for each Turán difference. The converse follows from an exact counterexample at $(r, n, k) = (6, 4, 2)$, derived here in two independent ways, and a monotone factorial ratio that gives counterexamples in row three for every $r \geq 7$. The same method proves log-concavity for the cycle triangles of orders three and four and for a continuous family of interpolating recurrences, and an asymptotic analysis explains why the elementary mechanism changes exactly between orders five and six. We also exhibit a precise obstruction to extending the method to the fifth-order cycle triangle; that separate conjectural case is not settled here. All arguments are given in ordinary mathematical form. Two independent verification toolchains accompany the report: a standard-library package that checks the polynomial identities exactly without a computer algebra system, and a SymPy checker of the same identities.

Scope and verification. This is an unrefereed research manuscript, not an independently certified publication. The selected subset-number assertion has a complete proof below; the claim is not based on a numerical search. The intended resolution is precisely the *log-concavity clause* of Conjecture 1.4(c) in [1]; it is not a resolution of that paper’s total-positivity, real-rootedness, or cycle-number conjectures. The originating preprint still states the selected assertion as a conjecture in the version accessed for this report. A targeted literature check did not locate a later resolution, but this does not certify priority. This manuscript has not undergone independent peer review or proof-assistant verification.

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Reading guide

The proof of the selected conjecture is in Sections 2–6. Section 3 isolates the local inequality; Section 4 contains the order-five certificate. A reviewer in a hurry should read the single identity (4.12), then the six-term display (4.18), which is the entire interior induction step in one equation. Sections 7–9 give the cycle-number extensions, an exact obstruction, and an asymptotic explanation. Sections 10–11 supply generating functions, sequence data, and the computational audit. The appendices collect the polynomial formulas, the deliberate redundancies retained from the two source drafts, a minimal symbolic check, and source-status details. Throughout, watch the distinct roles of the row index n , the column index k , the ground-set size $N = n + (r - 1)k$, and the abbreviation $t = n + (r - 1)k - 1$.

1 The problem and the results

Deb and Sokal introduce higher-order Stirling triangles and conjecture row log-concavity through order five in [1, Conjectures 1.3(c) and 1.4(c)]. The cycle-number cases $r = 3, 4, 5$ also appear as Problem 30.9 in the BCC30 problem collection [2]. We select the subset-number assertion because its recurrence has a particularly useful linear coefficient. The second-order instance is the Ward triangle, recorded as OEIS A134991 after removing the zero column [5].

Write $\left\{ \begin{smallmatrix} N \\ k \end{smallmatrix} \right\}_{\geq r}$, also written $B_r(N, k)$ below, for the number of set partitions of $[N]$ into k blocks of size at least r , and $\left[\begin{smallmatrix} N \\ k \end{smallmatrix} \right]_{\geq r}$ for the number of permutations of $[N]$ with k cycles of length at least r . Here $[N] = \{1, \dots, N\}$ and the empty structure has size zero. The arrays of interest are

$$S_{n,k}^{(r)} = \left\{ \begin{smallmatrix} n + (r-1)k \\ k \end{smallmatrix} \right\}_{\geq r}, \quad C_{n,k}^{(r)} = \left[\begin{smallmatrix} n + (r-1)k \\ k \end{smallmatrix} \right]_{\geq r}. \quad (1.1)$$

Their initial row is $(1, 0, 0, \dots)$, and entries outside $0 \leq k \leq n$ are zero. In every nonzero row, the positive entries are precisely those with $1 \leq k \leq n$.

Definition 1.1. A nonnegative sequence (a_k) is *log-concave* if $a_k^2 \geq a_{k-1}a_{k+1}$ at every relevant index. For a finite sequence we extend it by zeros. We say it is *strictly log-concave on the interior of its positive support* when the inequality is strict whenever all three entries are positive.

Theorem 1.2 (Sharp subset-number classification). *For a positive integer r , the following are equivalent:*

- (i) Every row of $S^{(r)}$ is log-concave.
- (ii) $1 \leq r \leq 5$.

For these five orders, and for every $n \geq 2$ and $1 \leq k \leq n-1$,

$$(S_{n,k}^{(r)})^2 > S_{n,k-1}^{(r)} S_{n,k+1}^{(r)}. \quad (1.2)$$

For $r = 6$ the first failing row is $n = 4$: its relevant entries are $(S_{4,1}^{(6)}, S_{4,2}^{(6)}, S_{4,3}^{(6)}) = (1, 4719, 23279256)$ and its log-concavity defect is -1010295 . For every $r \geq 7$ the first failing row is $n = 3$, with the failure at $k = 2$.

Theorem 1.3 (Cycle-number consequence). *Every row of $C^{(r)}$ is log-concave for $1 \leq r \leq 4$, with strict inequalities on the interior of the positive support. In particular, the log-concavity cases $r = 3, 4$ of Deb–Sokal’s Conjecture 1.3(c) hold, as do the cases $r = 3, 4$ of BCC30 Problem 30.9.*

These theorems concern the log-concavity clauses only. The same numbered conjectures in [1] contain other assertions; no claim about all their clauses follows from this manuscript. In particular, we do not settle the order-five cycle case, the general total positivity of the triangles, or their Hankel-total-positivity questions. Nor do the local inequalities proved here give total positivity of a row Toeplitz matrix: example 5.2 exhibits a strictly log-concave row whose polynomial has nonreal zeros, which already rules out the real-rootedness reading.

A crucial distinction is that n in (1.1) is *not* the cardinality of the underlying set. Moving right in a fixed row changes that cardinality by $r-1$. Thus a theorem about a fixed- N row of ordinary associated Stirling numbers would not, by itself, solve this problem. Likewise, these numbers must not be confused with the differently defined r -Stirling numbers, which require certain distinguished elements to lie in distinct blocks; compare the terminology in [4, Section 1].

1.1 What makes the proof work

The proof has three ingredients. First, a geometric rescaling removes the factorial denominator from the subset recurrence. Second, three elementary consequences of log-concavity reduce the next-row inequality to a quadratic form in two adjacent entries of the previous row. Finally, the triangular index restriction bounds its mixed coefficient. Orders three and four then follow from nonnegative coefficients. At order five the mixed coefficient can be negative — at $n = 3$, $k = 2$ it equals -4944 — so it must be controlled rather than discarded. The control is exact: a single identity, (4.12), writes twenty-five times the residual quadratic as a square plus two terms whose nonnegativity is exactly the triangular domain condition.

This is closely related to Sagan’s recurrence criterion [3, Theorem 1], whose nonnegative mixed-term hypothesis is sufficient for some of the lower-order cases; the same coefficient-sign condition is recorded as [4, Theorem 1.2]. [D8] Our local calculation retains the positive square terms and allows a negative mixed term. We do not claim that the general quadratic-form observation is historically new; the specific application and factorization are what establish the selected assertion here. Shankar’s principal power-of-affine-coefficients family is different from the falling-factorial weight used below.

1.2 Provenance of this document

This manuscript merges two independently produced research packages that proved the same theorem by the same four moves under different letters. The first, in the repository directory `docs/reports/log-concavity-and-unimodality/higher-order-stirling-rows`, was prepared on 19 September 2026; it proved the shared theorem by bounding the mixed coefficient $H_d(t, k)$ below by its value at the non-integral argument $k = t/(d + 1)$, using strict convexity of the falling factorial (lemma 4.1), and then completing the square against the factored discriminant $4A_4 - L_4^2$. It additionally proved the cycle cases $1 \leq r \leq 4$, the interpolating-family theorem, and the asymptotic explanation of the cutoff. The second, in `docs/reports/log-concavity-and-unimodality/stirling-subset-cutoff`, was prepared independently on 20 September 2026 and is recorded there as having been drafted with ChatGPT; it proved the shared theorem by the exact identity $H_4 - L_4 = \frac{96}{5}\beta(t)(t - 5k)$, which replaces the convexity argument by the immediate inequality $t - 5k = n - k - 1 \geq 0$, and folded the whole interior step into the single nonnegative-summand identity (4.12).

The present document keeps the first package as its spine and its notation, takes the second package’s identity as the primary fifth-order certificate, and retains both routes wherever they constitute independent checks. The twelve deliberate redundancies are tagged in the margin of the text as [D1]–[D12] and are listed with their purpose in section B.

2 Recurrences, normalization, and boundary values

Set $d = r - 1$, and define the falling-factorial polynomial

$$P_d(t) = \prod_{j=0}^{d-1} (t - j), \quad P_0(t) = 1. \quad (2.1)$$

The two recurrences in [1, Lemma 1.1] can be derived directly from the distinguished largest label, as follows.

Proposition 2.1. For $n \geq 1$, $1 \leq k \leq n$, and $t = n + dk - 1$,

$$S_{n,k}^{(r)} = \binom{t}{d} S_{n-1,k-1}^{(r)} + k S_{n-1,k}^{(r)} \quad (2.2)$$

$$C_{n,k}^{(r)} = P_d(t) C_{n-1,k-1}^{(r)} + t C_{n-1,k}^{(r)}. \quad (2.3)$$

Proof. Let $N = n + dk$. In a partition counted by $\left\{ \begin{smallmatrix} N \\ k \end{smallmatrix} \right\}_{\geq r}$, the block containing N either has exactly r elements or has more than r . In the first case choose its $r - 1$ other elements and partition the remaining $N - r$ elements into $k - 1$ admissible blocks. In the second case remove N ; there is an admissible partition of $N - 1$ elements and a choice of one of its k blocks in which to reinsert N . Hence

$$\left\{ \begin{smallmatrix} N \\ k \end{smallmatrix} \right\}_{\geq r} = \binom{N-1}{r-1} \left\{ \begin{smallmatrix} N-r \\ k-1 \end{smallmatrix} \right\}_{\geq r} + k \left\{ \begin{smallmatrix} N-1 \\ k \end{smallmatrix} \right\}_{\geq r}.$$

For permutations, a cycle of size exactly r has $(r - 1)!$ cyclic orderings once its elements have been chosen. Otherwise, removing N leaves a permutation into whose cycles N can be reinserted after any of the $N - 1$ remaining labels. The corresponding weights are therefore $(r - 1)! \binom{N-1}{r-1}$ and $N - 1$. Finally, $N - r = (n - 1) + d(k - 1)$ and $N - 1 = (n - 1) + dk$, giving the stated recurrences. $\square$

The argument includes $r = 1$, where the first case means that N is a singleton. Since $B_r(N, k)$ vanishes for $0 < N < rk$, the substitution $N = n + (r - 1)k$ makes the support exactly $0 \leq k \leq n$: for $1 \leq k \leq n$ one may take $k - 1$ blocks of size r and one block of size $r + n - k$, so

$$S_{n,0}^{(r)} = 0, \quad S_{n,1}^{(r)} = 1, \quad S_{n,k}^{(r)} > 0 \quad (1 \leq k \leq n) \quad (2.4)$$

for $n \geq 1$.

For the subset array define

$$T_{n,k}^{(r)} = (d!)^k S_{n,k}^{(r)}. \quad (2.5)$$

Then (2.2) becomes

$$T_{n,k}^{(r)} = P_d(n + dk - 1) T_{n-1,k-1}^{(r)} + k T_{n-1,k}^{(r)} \quad (2.6)$$

an exact equivalence because $P_d(t) = d! \binom{t}{d}$ and the factor $(d!)^k$ supplies one extra $d!$ in the first summand. This normalization leaves log-concavity unchanged, since

$$(T_{n,k}^{(r)})^2 - T_{n,k-1}^{(r)} T_{n,k+1}^{(r)} = (d!)^{2k} [(S_{n,k}^{(r)})^2 - S_{n,k-1}^{(r)} S_{n,k+1}^{(r)}]. \quad (2.7)$$

The cycle array already has the same first weight $P_d(t)$; only its second weight differs.

For later checks, direct counting gives, for $n \geq 1$,

$$S_{n,1}^{(r)} = 1, \quad S_{n,n}^{(r)} = \frac{(rn)!}{n!(r!)^n}, \quad (2.8)$$

$$C_{n,1}^{(r)} = (n + r - 2)!, \quad C_{n,n}^{(r)} = \frac{(rn)!}{n!r^n}. \quad (2.9)$$

In particular $C_{1,1}^{(r)} = (r - 1)!$; for $r > 2$ this is not one. These values also make the support assertions immediate from the recurrences. The rows $n = 0, 1$ carry no internal inequality, and the only internal index of row $n = 2$ is $k = 1$, handled in section 4.

2.1 An independent counting formula

A second, definition-level formula will be used for an independent check of the small entries. A partition of $[N]$ with exactly m_j blocks of size j is counted by

$$\frac{N!}{\prod_{j \geq r} (j!)^{m_j} m_j!}, \quad \sum_{j \geq r} m_j = k, \quad \sum_{j \geq r} j m_j = N : \quad (2.10)$$

order all N elements, cut them into blocks of the prescribed sizes, and divide out the order inside each block and the permutations of equally sized blocks. Summing (2.10) over admissible block-size profiles gives $B_r(N, k)$. This route uses neither the recurrence nor the normalization, and it is different from the generating-function route of section 10; both are retained. [D4]

3 A sharp local quadratic-form criterion

Let (x_j) be a nonnegative log-concave sequence with no internal zeros. At a fixed index k suppose $x_{k-1} > 0$ and $x_k > 0$, and put

$$w = x_{k-2}, \quad u = x_{k-1}, \quad v = x_k, \quad z = x_{k+1}.$$

Lemma 3.1 (Three slacks). *With the preceding notation,*

$$wv \leq u^2, \quad uz \leq v^2, \quad wz \leq uv. \quad (3.1)$$

Proof. The first two are log-concavity at $k-1$ and k . For the third, if $wz = 0$ the claim is immediate; otherwise the interval-support condition makes all four entries positive, and multiplying the first two inequalities and dividing by $uv > 0$ gives $wz \leq uv$. $\square$

Consider nonnegative coefficients and the local transform

$$y_j = a_j x_{j-1} + b_j x_j \quad (j = k-1, k, k+1). \quad (3.2)$$

Use the abbreviations $a = a_k$, $a_- = a_{k-1}$, $a_+ = a_{k+1}$ and similarly for b . Define

$$A = a^2 - a_- a_+, \quad B = b^2 - b_- b_+, \quad H = 2ab - a_- b_+ - b_- a_+. \quad (3.3)$$

Lemma 3.2 (Exact slack decomposition). *With the preceding notation,*

$$\begin{aligned} y_k^2 - y_{k-1} y_{k+1} &= Au^2 + Huv + Bv^2 \\ &\quad + a_- a_+ (u^2 - wv) + b_- b_+ (v^2 - uz) + a_- b_+ (uv - wz). \end{aligned} \quad (3.4)$$

Consequently,

$$y_k^2 - y_{k-1} y_{k+1} \geq Au^2 + Huv + Bv^2. \quad (3.5)$$

Proof. Expand $(au + bv)^2 - (a_- w + b_- u)(a_+ v + b_+ z)$ and collect terms. This gives (3.4) exactly. Each of the three remaining brackets is nonnegative by lemma 3.1, and each multiplier is nonnegative. $\square$

Proposition 3.3 (Local preservation criterion). *The quadratic form $Au^2 + Huv + Bv^2$ is nonnegative for every $u, v > 0$ if and only if*

$$A \geq 0, \quad B \geq 0, \quad H \geq -2\sqrt{AB}. \quad (3.6)$$

Thus these conditions imply the next-row log-concavity inequality. They are also necessary if the transform is required to satisfy that inequality for every positive log-concave four-term input.

Proof. If (3.6) holds, write

$$Au^2 + Huv + Bv^2 = (\sqrt{A}u - \sqrt{B}v)^2 + (H + 2\sqrt{AB})uv \geq 0.$$

Conversely, sending v/u to zero or infinity forces $A \geq 0$ and $B \geq 0$. For $A, B > 0$, choosing $v/u = \sqrt{A/B}$ forces the third condition. If $A = 0$ or $B = 0$, the same condition, now $H \geq 0$, follows by taking an appropriately small or large positive ratio.

For necessity as a preservation criterion, take the four input values

$$(w, u, v, z) = (u^2/v, u, v, v^2/u).$$

They form a positive geometric progression, so all three slacks in (3.4) vanish. The next-row difference is exactly the quadratic form. Extending this four-term sequence by zeros produces no difficulty at the support boundaries. $\square$

Only nonnegativity on the positive quadrant is needed, not positive semidefiniteness on all of $\mathbb{R}^2$. In matrix terminology this is copositivity of a 2×2 symmetric matrix. For exact computation the square root in (3.6) can be avoided entirely: require $A, B \geq 0$, and then require either $H \geq 0$ or $H^2 \leq 4AB$. For the hard subset case we will use the stronger, more easily certified condition $4AB - H^2 > 0$ after replacing H by a suitable lower bound — and in fact even that check is replaced, in lemma 4.3, by an identity with visibly nonnegative terms.

Remark 3.4 (Why a negative mixed term is not fatal). Sagan's sufficient condition [3, Theorem 1], and equally the coefficient-sign criterion [4, Theorem 1.2], correspond to $A, B, H \geq 0$. If $H < 0$, rejecting the recurrence at once loses the positive contributions Au^2 and Bv^2 . The exact square-completion threshold in (3.6) is what allows us to handle the fifth-order subset array and the fourth-order cycle array. [D8]

4 Proof of subset log-concavity through order five

Assume row $n - 1$ of $T^{(r)}$ is log-concave. The index $k = 1$ is immediate from (2.4) and (2.7): since $S_{n,0}^{(r)} = 0$ and $S_{n,1}^{(r)} = 1$, the normalized defect at $k = 1$ equals $(d!)^2 > 0$. For the remaining internal indices fix

$$n \geq 3, \quad 2 \leq k \leq n - 1, \quad d = r - 1, \quad t = n + dk - 1. \quad (4.1)$$

For (2.6), the local weights are

$$a = P_d(t), \quad a_- = P_d(t - d), \quad a_+ = P_d(t + d), \quad b = k, \quad b_- = k - 1, \quad b_+ = k + 1.$$

Notice that the step in t between *adjacent columns of one row* is d , not $d + 1$. The latter number enters a different way: since $n \geq k + 1$,

$$t = (d + 1)k + (n - k - 1) \geq (d + 1)k, \quad k \leq \frac{t}{d + 1}, \quad t \geq 2(d + 1). \quad (4.2)$$

Treating t and k as unconstrained independent parameters would discard exactly the information the fifth-order argument needs.

Let

$$A_d(t) = P_d(t)^2 - P_d(t - d)P_d(t + d), \quad (4.3)$$

$$H_d(t, k) = 2kP_d(t) - (k + 1)P_d(t - d) - (k - 1)P_d(t + d). \quad (4.4)$$

The remaining quadratic coefficient is simply $B = 1$.

4.1 Two elementary properties of the falling factorial

Lemma 4.1. *For $d \geq 1$ and $t > 2d - 1$, $A_d(t) > 0$. For $d \geq 2$ in the same domain,*

$$\Delta_d(t) := P_d(t - d) + P_d(t + d) - 2P_d(t) > 0.$$

For $d = 0, 1$, the centered second difference is zero.

Proof. For $d \geq 1$,

$$P_d(t - d)P_d(t + d) = \prod_{j=0}^{d-1} ((t - j)^2 - d^2).$$

Every factor is positive and strictly smaller than $(t - j)^2$, because $t > 2d - 1$. Comparing products proves $A_d(t) > 0$.

For $d \geq 2$, $P_d''(s) > 0$ whenever $s > d - 1$: differentiate the product twice, obtaining a sum of positive products of the remaining $d - 2$ factors. Thus P_d is strictly convex on the interval $[t - d, t + d]$, and its centered second difference is positive. The polynomials $P_0 = 1$ and $P_1 = t$ have zero centered second difference. $\square$

This lemma is used three times: for $A_d > 0$ in every order treated here, for the interpolating family of theorem 7.1, and for the asymptotics of section 9. Rewriting the mixed coefficient as

$$H_d(t, k) = P_d(t + d) - P_d(t - d) - k\Delta_d(t) \quad (4.5)$$

shows that for fixed t it is affine and nonincreasing in k . Hence (4.2) yields the lower bound

$$H_d(t, k) \geq L_d(t) := H_d\left(t, \frac{t}{d+1}\right). \quad (4.6)$$

The argument $t/(d+1)$ need not be an integer; $H_d(t, k)$ is affine in k , so this substitution is legitimate.

Remark 4.2 (Two routes to the mixed-coefficient bound). At $d = 4$ the bound (4.6) is not needed in this form: the exact identity (4.13) below gives $H_4 - L_4 = \frac{96}{5}\beta(t)(t - 5k)$, so $H_4 \geq L_4$ follows from $t - 5k = n - k - 1 \geq 0$ and $\beta > 0$ with no appeal to convexity at all. That identity is the route used in the proof. The convexity route is retained because it is the only one of the two that survives to general d : theorem 7.1 and section 9 both depend on $\Delta_d(t) > 0$. [D2]

4.2 The orders $r = 1, 2, 3, 4$

Direct expansion gives

r	d	$L_d(t)$
1	0	0
2	1	2
3	2	$\frac{4}{3}(4t - 3)$
4	3	$\frac{3}{2}(3t^2 - 15t + 44)$

For $d = 0$, $A_0 = 0$, $H_0 = 0$, and the quadratic lower bound is $v^2 > 0$. For $d = 1$, $A_1 = 1$ and $H_1 = 2$, so it is $(u + v)^2 > 0$. For $d = 2$, $H_2 = 4(2t - 2k - 1)$ and $k \leq t/3$ give $H_2 \geq \frac{4}{3}(4t - 3) > 0$ because $t \geq 6$. For $d = 3$, positivity follows even without the domain restriction from

$$3t^2 - 15t + 44 = 3(t - 5/2)^2 + 101/4 > 0,$$

equivalently from the negative discriminant $225 - 528 < 0$. In the latter two cases $A_d > 0$ by lemma 4.1, so the quadratic form is strictly positive for $u, v > 0$.

4.3 The order $r = 5$

Here $d = 4$, $t = n + 4k - 1 \geq 5k \geq 10$, and

$$P_4(t) = t(t - 1)(t - 2)(t - 3), \quad (4.7)$$

$$A_4(t) = 32(2t^6 - 18t^5 + 17t^4 + 168t^3 - 55t^2 - 834t - 630), \quad (4.8)$$

$$H_4(t, k) = 16(-12kt^2 + 36kt - 54k + 2t^3 - 9t^2 + 43t - 51), \quad (4.9)$$

$$L_4(t) = -\frac{16}{5}(2t^3 + 9t^2 - 161t + 255). \quad (4.10)$$

Unlike the lower orders, H_4 is not always nonnegative: at $n = 3$, $k = 2$, so $t = 10$, it equals -4944 . The coefficient-sign criterion of remark 3.4 therefore cannot prove this case, and the quadratic form must be used in full.

Introduce the three small polynomials

$$\alpha(t) = 2t^3 + 9t^2 - 161t + 255, \quad \beta(t) = 2t^2 - 6t + 9, \quad \gamma(t) = 7t^3 - 31t^2 - 126t + 1080, \quad (4.11)$$

so that $L_4(t) = -\frac{16}{5}\alpha(t)$. (These are the polynomials written A, B, C in the second source draft; they are renamed here to avoid a collision with the coefficients A_d and B of (3.3).)

Lemma 4.3 (Integer square certificate). *For every real t, k, u, v ,*

$$\boxed{25(A_4(t)u^2 + H_4(t, k)uv + v^2) = (5v - 8\alpha(t)u)^2 + 96(t - 5)\beta(t)\gamma(t)u^2 + 480\beta(t)(t - 5k)uv.} \quad (4.12)$$

If $t \geq 10$, $t \geq 5k$, and $u, v \geq 0$, the right-hand side is nonnegative, and it is strictly positive whenever $u > 0$.

Proof. The identity is verified by multiplying out both sides. A less bulky verification uses the two exact identities

$$H_4(t, k) - L_4(t) = \frac{96}{5}\beta(t)(t - 5k), \quad (4.13)$$

$$4A_4(t) - L_4(t)^2 = \frac{384}{25}(t - 5)\beta(t)\gamma(t). \quad (4.14)$$

Indeed, (4.5) shows that H_4 is affine in k with slope $-\Delta_4(t) = -96\beta(t)$ and that $L_4 = H_4(t, t/5)$, which is exactly (4.13). Completing the square in v gives

$$A_4u^2 + L_4uv + v^2 = \left(v + \frac{L_4}{2}u\right)^2 + \frac{4A_4 - L_4^2}{4}u^2,$$

and multiplying by 25, using $\frac{5}{2}L_4 = -8\alpha$ and (4.14), turns this into the first two summands of (4.12). Adding $25(H_4 - L_4)uv$ and applying (4.13) supplies the third. Equivalently, (4.12) is the conjunction of the three coefficientwise identities

$$25A_4 = 64\alpha^2 + 96(t-5)\beta\gamma, \quad 25H_4 = -80\alpha + 480\beta(t-5k), \quad 25 = 25,$$

each of which is a polynomial identity in t alone once the affine k -dependence is separated. All of them are checked by exact coefficient comparison in the accompanying `certificates.py`, which uses no computer algebra system, and again in `code/verify_certificates.py`, which uses SymPy. [D5]

For the signs, first

$$\beta(t) = 2\left(t - \frac{3}{2}\right)^2 + \frac{9}{2} > 0 \quad (4.15)$$

for all real t , and, writing $t = 10 + s$ with $s \geq 0$,

$$\gamma(10 + s) = 7s^3 + 179s^2 + 1354s + 3720 > 0. \quad (4.16)$$

Also $t - 5 > 0$ and $t - 5k \geq 0$ on the domain. Hence all three summands of (4.12) are nonnegative for $u, v \geq 0$, and the middle one is strictly positive when $u > 0$. $\square$

Two further forms of the same certificate are kept, because each is checkable in a way the others are not. [D1] The two-step form is the factored discriminant (4.14) together with the completed square

$$\begin{aligned} A_4(t)u^2 + H_4(t, k)uv + v^2 &\geq A_4(t)u^2 + L_4(t)uv + v^2 \\ &= \left(v + \frac{L_4(t)}{2}u\right)^2 + \frac{4A_4(t) - L_4(t)^2}{4}u^2 > 0; \end{aligned} \quad (4.17)$$

this is the form that generalizes, unchanged in shape, to the cycle coefficients of section 7 and to arbitrary d in section 9. The fully expanded coefficient form of (4.14) is recorded in section A; it is the only version that a reader can check entirely by hand.

4.4 The interior induction step in one display

Collecting lemma 3.2 and lemma 4.3 gives a single equation that is the whole interior step of the induction. [D1]

Proposition 4.4 (Six-term defect certificate). *Let $r = 5$, let $n \geq 3$ and $2 \leq k \leq n - 1$, and put*

$$a_j = T_{n-1,j}^{(5)}, \quad b_j = T_{n,j}^{(5)}, \quad t = n + 4k - 1, \quad u = a_{k-1}, \quad v = a_k.$$

Then, with $P_4, \alpha, \beta, \gamma$ as above,

$$\begin{aligned} 25(b_k^2 - b_{k-1}b_{k+1}) &= 25P_4(t-4)P_4(t+4)(a_{k-1}^2 - a_{k-2}a_k) \\ &\quad + 25(k^2 - 1)(a_k^2 - a_{k-1}a_{k+1}) \\ &\quad + 25P_4(t-4)(k+1)(a_{k-1}a_k - a_{k-2}a_{k+1}) \\ &\quad + (5v - 8\alpha(t)u)^2 \\ &\quad + 96(t-5)\beta(t)\gamma(t)u^2 \\ &\quad + 480\beta(t)(n-k-1)uv. \end{aligned} \quad (4.18)$$

Proof. Multiply (3.4) by 25, with the weights $a_-a_+ = P_4(t-4)P_4(t+4)$, $b_-b_+ = (k-1)(k+1)$ and $a_-b_+ = P_4(t-4)(k+1)$, and replace the resulting $25(A_4u^2 + H_4uv + v^2)$ by the right-hand side of (4.12), using $t - 5k = n - k - 1$. $\square$

The signs of the six summands require, in order: log-concavity of row $n - 1$ at $k - 1$; log-concavity at k ; lemma 3.1; that a square is nonnegative; $t \geq 10$ together with (4.15) and (4.16); and $n - k - 1 \geq 0$. The fifth summand is strictly positive because $u > 0$.

4.5 Completing the induction, including the edges

The rows $n = 0, 1$ have no internal inequality, and row $n = 2$ has only the index $k = 1$ treated above. Assume row $n - 1$ is log-concave. For $2 \leq k \leq n - 1$, both $u = T_{n-1,k-1}^{(r)}$ and $v = T_{n-1,k}^{(r)}$ are positive. At $k = 2$, $w = 0$; at $k = n - 1$, $z = 0$. Neither zero causes a problem in lemma 3.2. The preceding calculations show that its quadratic lower bound is strictly positive for every order $r \leq 5$; for $r = 5$ this is lemma 4.3, or in one line (4.18). Hence all the positive-support interior inequalities in row n are strict. At either endpoint of the positive support one neighboring entry is zero, and outside the support the inequalities are trivial. This completes the induction for $T^{(r)}$. Equation (2.7) transfers the result to $S^{(r)}$.

Three features of this induction deserve emphasis. Nothing in the argument requires strict log-concavity of the input row: ordinary log-concavity, the interval support condition, and $u > 0$ already suffice, so the induction *creates* strictness rather than silently assuming it. The last internal index $k = n - 1$ is genuinely included, since there $a_{k+1} = 0$ and $t - 5k = 0$, both of which the certificate allows. And the smallest genuine case $n = 3$, $k = 2$, $t = 10$ is covered without an asymptotic exception.

We have proved the sufficiency direction of theorem 1.2, including the strict inequalities, without using real-rootedness or a finite verification bound.

5 An explicit fifth-order Turán inequality

The certificate gives more than the sign of a difference. Let

$$\Theta(t) = (t - 5)\beta(t)\gamma(t) = (t - 5)(2t^2 - 6t + 9)(7t^3 - 31t^2 - 126t + 1080). \quad (5.1)$$

Corollary 5.1. For $n \geq 3$, $2 \leq k \leq n - 1$, and $t = n + 4k - 1$,

$$(S_{n,k}^{(5)})^2 - S_{n,k-1}^{(5)} S_{n,k+1}^{(5)} \geq \frac{\Theta(t)}{150} (S_{n-1,k-1}^{(5)})^2 > 0. \quad (5.2)$$

Proof. Drop every summand of (4.18) except the fifth. The difference for the normalized row is then at least $(96/25)\Theta(t)(T_{n-1,k-1}^{(5)})^2$. Now divide by 24^{2k} and use $T_{n-1,k-1}^{(5)} = 24^{k-1}S_{n-1,k-1}^{(5)}$. The remaining constant is $96/(25 \cdot 24^2) = 1/150$. $\square$

For instance, the nonzero portion of row three is

$$(S_{3,1}^{(5)}, S_{3,2}^{(5)}, S_{3,3}^{(5)}) = (1, 462, 126126).$$

Its middle difference is $462^2 - 126126 = 87318$. The lower bound in (5.2), with $t = 10$, is 18476. It is not asserted to be optimal; its point is to make strict positivity uniform and explicit. At that same smallest index the scalar coefficients are

$$A_4(10) = 16752960, \quad H_4(10, 2) = -4944, \quad \alpha(10) = 1545, \quad \beta(10) = 149, \quad \gamma(10) = 3720,$$

so that $4A_4 - H_4^2 = 42568704 > 0$ although $H_4 < 0$: this audits the smallest nontrivial fifth-order case and shows concretely how much the positive square terms retain.

5.1 Shape consequences

In a positive row, (1.2) is equivalent to strict decrease of successive ratios:

$$\frac{S_{n,2}^{(r)}}{S_{n,1}^{(r)}} > \frac{S_{n,3}^{(r)}}{S_{n,2}^{(r)}} > \dots > \frac{S_{n,n}^{(r)}}{S_{n,n-1}^{(r)}}. \quad (5.3)$$

Therefore every such row increases until its successive ratios cross 1 and then decreases; it is unimodal, with either a unique maximum or two adjacent equal maxima, since strict log-concavity does not exclude the latter. Multiplying entries by q^k , for any $q > 0$, preserves the theorem. Normalizing these weighted entries by their sum gives a log-concave probability mass function on the column index. This is a distribution over a row of the shifted array, not over partitions of a single fixed set size.

The same ratio comparison yields a formally stronger statement, which does not follow from the unimodality remark and does not imply it: [D11]

$$S_{n,i}^{(r)} S_{n,j}^{(r)} > S_{n,i-1}^{(r)} S_{n,j+1}^{(r)} \quad (2 \leq i \leq j \leq n-1). \quad (5.4)$$

Indeed, compare the ratio at i with the ratio at $j+1$ in (5.3) and multiply by the positive denominators. Cases involving a zero outside the support satisfy the corresponding weak inequality. No separate combinatorial injection is needed.

5.2 Log-concavity without real-rootedness

Example 5.2. For $r = 5$ and $n = 3$ the row polynomial is

$$x + 462x^2 + 126126x^3 = x(1 + 462x + 126126x^2).$$

The row is strictly log-concave, since $462^2 - 126126 = 87318 > 0$, yet the quadratic factor has discriminant

$$462^2 - 4 \cdot 126126 = -291060 < 0,$$

so its roots are not real.

This example is a scope fence. [D12] Any argument that tried to prove real-rootedness first would be attempting a statement that is false for this family. It also shows that the local inequalities established here cannot yield total positivity of the triangle or nonnegativity of all minors of a row Toeplitz matrix. The complementary fence is the result-to-source map in section D, which records exactly which clause of which conjecture is and is not settled.

6 Why order five is the exact cutoff

6.1 The sixth-order counterexample

For $r = 6$ the first relevant rows are

$$\begin{aligned} (S_{2,k}^{(6)})_{k=0}^2 &= (0, 1, 462), \\ (S_{3,k}^{(6)})_{k=0}^3 &= (0, 1, 1716, 2858856), \\ (S_{4,k}^{(6)})_{k=0}^4 &= (0, 1, 4719, 23279256, 96197645544). \end{aligned}$$

Row three passes, because $1716^2 - 2858856 = 85800 > 0$. The two entries that decide row four can be obtained in two independent ways, and both are recorded here because this is the single load-bearing numerical fact of the whole converse. [D3] By block-size pattern,

$$S_{4,2}^{(6)} = \binom{14}{6} + \frac{1}{2} \binom{14}{7} = 4719,$$

$$S_{4,3}^{(6)} = \frac{19!}{2! (6!)^2 7!} = 23279256,$$

the second entry counting the patterns (6, 8) and (7, 7) and the third the pattern (6, 6, 7). By the recurrence (2.2),

$$S_{4,2}^{(6)} = \binom{13}{5} + 2 \cdot 1716 = 4719,$$

$$S_{4,3}^{(6)} = \binom{18}{5} \cdot 1716 + 3 \cdot 2858856 = 23279256.$$

Consequently,

$$(S_{4,2}^{(6)})^2 - S_{4,1}^{(6)} S_{4,3}^{(6)} = 22268961 - 23279256 = -1010295 < 0. \quad (6.1)$$

Rows zero and one have no internal test, and the only test for row two has a zero left neighbour. Together with the row-three calculation, this proves that row four is the first failure for $r = 6$.

6.2 A row-three counterexample for every $r \geq 7$

The nonzero part of row three has a uniform closed form:

$$S_{3,1}^{(r)} = 1, \quad S_{3,2}^{(r)} = \binom{2r+1}{r}, \quad S_{3,3}^{(r)} = \frac{(3r)!}{6(r!)^3}. \quad (6.2)$$

The two-block case has block sizes $r, r+1$, and the three-block case has three blocks of size r , as in (2.8). Define the ratio that decides log-concavity in this row:

$$F(r) = \frac{\binom{2r+1}{r}^2}{(3r)!/[6(r!)^3]}, \quad R(r) = \frac{1}{F(r)} = \frac{(3r)!(r+1)!^2}{6r!(2r+1)!^2}. \quad (6.3)$$

The row-three test fails exactly when $F(r) < 1$, equivalently when $R(r) > 1$. Both presentations are kept, since each is a transcription check on the other. [D9] Cancellation of consecutive factorials gives

$$\frac{F(r+1)}{F(r)} = \frac{4(r+1)^2(2r+3)^2}{3(r+2)^2(3r+2)(3r+1)}, \quad \frac{R(r+1)}{R(r)} = \frac{3(3r+1)(3r+2)(r+2)^2}{4(r+1)^2(2r+3)^2}. \quad (6.4)$$

In either form, the denominator of $F(r+1)/F(r)$ minus its numerator is

$$11r^4 + 55r^3 + 74r^2 + 12r - 12 > 0 \quad (r \geq 1). \quad (6.5)$$

For a coefficientwise positivity certificate, substitute $r = 1 + w$; the polynomial becomes

$$11w^4 + 99w^3 + 305w^2 + 369w + 140.$$

Thus $F(r)$ is strictly decreasing, and $R(r)$ strictly increasing, on the positive integers. At $r = 7$,

$$\binom{15}{7}^2 = 6435^2 = 41409225 < 66512160 = \frac{21!}{6(7!)^3}.$$

Therefore $F(r) < 1$ for every $r \geq 7$, proving a failure in row three. No earlier row can fail because it has at most two positive entries. Together with (6.1), this proves necessity and completes theorem 1.2.

6.3 The third row at the threshold

r	$S_{3,2}^{(r)}$	$S_{3,3}^{(r)}$	$(S_{3,2}^{(r)})^2 - S_{3,3}^{(r)}$
1	3	1	8
2	10	15	85
3	35	280	945
4	126	5775	10101
5	462	126126	87318
6	1716	2858856	85800
7	6435	66512160	-25102935
8	24310	1577585295	-986609195

Table 1: Exact third-row data; $S_{3,1}^{(r)} = 1$ for every r . The order-six line is the reason the converse needs row four.

table 1 makes the structure of the converse visible. The order-six row three still passes, and only its row four fails. The higher-order obstruction is therefore not a uniform single-row phenomenon beginning at order six: an argument that checked only row three would locate the cutoff at the wrong place.

7 Cycle triangles and an interpolating family

For the cycle recurrence, the first quadratic coefficient is the same $A_d(t)$, but

$$b = t, \quad b_- = t - d, \quad b_+ = t + d, \quad B = d^2.$$

The mixed coefficient is

$$M_d(t) = 2tP_d(t) - (t + d)P_d(t - d) - (t - d)P_d(t + d). \quad (7.1)$$

For $d = 1, 2, 3$ it is

$$M_1(t) = 2, \quad (7.2)$$

$$M_2(t) = 8(t - 1), \quad (7.3)$$

$$M_3(t) = -18(3t - 11). \quad (7.4)$$

The first two cases have a positive mixed coefficient. For $d = 3$, use

$$4 \cdot 9A_3(t) - M_3(t)^2 = 972(t - 3)^2(t - 1)(t + 3) > 0 \quad (t \geq 8). \quad (7.5)$$

Since $A_3(t) > 0$ and $B = 9$, the quadratic form is positive definite. This proves strict log-concavity for orders $r = 2, 3, 4$ by the same induction and boundary argument as in section 4. Note that the proof here is the completed-square form (4.17), not the single-identity form (4.12): the two-step route is what transfers to a different second weight.

For completeness, ordinary cycle numbers ($r = 1, d = 0$) can also be handled without a real-rootedness theorem. The recurrence has $a = 1$ and $b = n - 1$, independent of k , so $A = B = M = 0$. The first slack in (3.4) nevertheless supplies strictness. For $k = 2, w = 0$ and $u > 0$, hence $u^2 - wv > 0$. For $k > 2$, strict log-concavity in row $n - 1$ at index $k - 1$ gives the same inequality. The base row with two positive entries is automatic. This completes theorem 1.3.

7.1 A continuous extension

The subset and cycle results of orders at most four fit into a single family. The coefficients below need not be integers.

Theorem 7.1. *Let $1 \leq d \leq 3$, let $\rho, \sigma \geq 0$ with $\rho + \sigma > 0$, and define an array U with initial row $U_{0,k} = \delta_{0k}$, zero entries outside $0 \leq k \leq n$, and recurrence*

$$U_{n,k} = P_d(n + dk - 1)U_{n-1,k-1} + [\rho(n + dk - 1) + \sigma k]U_{n-1,k} \quad (7.6)$$

for $1 \leq k \leq n$, with $U_{n,0} = 0$ for $n \geq 1$. Every row is strictly log-concave on the interior of its positive support.

Proof. All entries on $1 \leq k \leq n$ are positive. At a fixed interior index, write $t = n + dk - 1$. The first quadratic coefficient remains $A_d(t)$. Since the second weight is affine in k with slope $\rho d + \sigma$,

$$B = (\rho d + \sigma)^2, \quad H = \rho M_d(t) + \sigma H_d(t, k).$$

Our preceding calculations give

$$M_d(t) > -2d\sqrt{A_d(t)}, \quad H_d(t, k) > -2\sqrt{A_d(t)}$$

for $d = 1, 2, 3$ in the triangular interior; the second bound uses (4.6), hence lemma 4.1. Multiplying by ρ, σ and adding yields

$$H > -2(\rho d + \sigma)\sqrt{A_d(t)} = -2\sqrt{A_d(t)}B.$$

The local criterion proves strict positivity of the quadratic form, and induction completes the proof. $\square$

The choice $(\rho, \sigma) = (0, 1)$ is the normalized subset array, and $(1, 0)$ is the cycle array. This theorem is a genuine parameter extension, not a convex-combination assertion about already computed rows: the parameters are inserted in the recurrence at every step.

8 A rigorous obstruction in the fifth-order cycle case

The preceding method does not automatically settle the remaining cycle case. In fact, universal preservation of log-concavity is false for its row operator, and we record two independent witnesses of this. [D7] At $d = 4$,

$$M_4(t) = -32(t - 2)(2t^2 + 4t - 51), \quad (8.1)$$

and the algebraic witness is the factorization

$$\boxed{64A_4(t) - M_4(t)^2 = -9216(t - 4)(t + 4)(2t - 9)\beta(t)}. \quad (8.2)$$

For $t \geq 10$ the right-hand side is negative and $M_4(t) < 0$. Thus the necessary condition of proposition 3.3 fails.

The numerical witness is an exact input that also respects the support of a preceding triangular row. Let $n = 5$, $q = 7350$, and take

$$(x_0, x_1, x_2, x_3, x_4, x_5) = (0, 1, q, q^2, q^3, 0).$$

It is log-concave. Apply the fifth-order cycle row operator

$$y_j = P_4(4 + 4j)x_{j-1} + (4 + 4j)x_j \quad (1 \leq j \leq 5).$$

The three relevant output entries are

$$y_2 = 100080, \quad y_3 = 1185408000, \quad y_4 = 14223043800000, \quad (8.3)$$

and their Turán difference is

$$y_3^2 - y_2y_4 = -18250097040000000 < 0. \quad (8.4)$$

The choice of q is not arbitrary. At this index $t = 16$, the quadratic form $A_4u^2 + M_4uv + 16v^2$ is minimized as a function of v/u at

$$\frac{v}{u} = -\frac{M_4(16)}{32} = (16 - 2)(2 \cdot 16^2 + 4 \cdot 16 - 51) = 7350.$$

The geometric input makes every slack zero and realizes the negative quadratic value exactly.

What this does and does not show. The artificial input is not a row of $C^{(5)}$. Equation (8.4), like the factorization (8.2), disproves only the claim that this operator preserves *all* log-concave inputs. Rows generated by one particular recurrence can satisfy additional inequalities that arbitrary log-concave inputs do not. Neither statement disproves log-concavity of the actual fifth-order Stirling cycle rows, and no resolution of that case — Deb–Sokal Conjecture 1.3(c) for $r = 5$, equivalently BCC30 Problem 30.9 for $r = 5$ — is claimed here. What (8.2) locates is the exact point at which the present universal-preservation argument ceases to apply.

There is a concrete next invariant to investigate. For the actual previous row, retain the three nonnegative slacks in (3.4) and seek a lower bound on their weighted sum that dominates the possible negative part of $A_4u^2 + M_4uv + 16v^2$. Alternatively, prove a restriction on the actual adjacent ratio v/u that excludes the dangerous interval between the two positive roots of

$$16q^2 + M_4(t)q + A_4(t) = 0.$$

Either approach would use information specific to the array, which ordinary row log-concavity alone cannot supply. These are proposed extensions, not additional proved statements.

9 A leading-order explanation of the cutoff

The exact proof is polynomial, but its structure has a useful asymptotic interpretation. Fix $d \geq 1$ and let $t \rightarrow \infty$. Since $P_d(t) = t^d + O(t^{d-1})$, centered Taylor expansions give

$$P_d(t + d) - P_d(t - d) = 2d^2t^{d-1} + O(t^{d-2}), \quad (9.1)$$

$$\Delta_d(t) = d^3(d - 1)t^{d-2} + O(t^{d-3}). \quad (9.2)$$

For $d = 1$ the centered second difference is exactly zero, as this formula permits. Comparing the leading terms in $P_d^2 - P_d(t - d)P_d(t + d)$ gives

$$A_d(t) = d^3t^{2d-2} + O(t^{2d-3}). \quad (9.3)$$

For example, this also follows from $\log P_d(t-d) + \log P_d(t+d) - 2\log P_d(t) = -d^3/t^2 + O(t^{-3})$. Combining (9.1)–(9.2) with $L_d = P_d(t+d) - P_d(t-d) - t\Delta_d/(d+1)$ yields

$$L_d(t) = \frac{d^2(-d^2 + 3d + 2)}{d+1} t^{d-1} + O(t^{d-2}). \quad (9.4)$$

Hence the normalized mixed coefficient has limit

$$\frac{L_d(t)}{2\sqrt{A_d(t)}} \longrightarrow \frac{\sqrt{d}(-d^2 + 3d + 2)}{2(d+1)}. \quad (9.5)$$

At $d = 4$ this is $-2/5$, safely above the local threshold -1 . At $d = 5$ it is $-2\sqrt{5}/3 < -1$. Thus the elementary preservation mechanism naturally changes between orders five and six. The exact counterexamples in section 6, rather than this asymptotic observation alone, prove the cutoff for the actual arrays.

For cycles, (7.1) can be written as $M_d = d(P_d(t+d) - P_d(t-d)) - t\Delta_d$. Therefore

$$M_d(t) = d^3(3-d)t^{d-1} + O(t^{d-2}), \quad \frac{M_d(t)}{2d\sqrt{A_d(t)}} \longrightarrow \frac{\sqrt{d}(3-d)}{2}. \quad (9.6)$$

At $d = 4$ the latter limit is exactly -1 , on the boundary. The leading terms of $64A_4$ and M_4^2 cancel; their difference begins with $-36864t^5$. This explains why the fifth-order cycle operator is more delicate than its subset counterpart, and why keeping lower-order terms is essential.

10 Generating functions and reproducible sequence data

10.1 An independent coefficient formula

Define formal power series

$$f_r(z) = \sum_{j \geq 0} \frac{z^j}{(r+j)!}, \quad g_r(z) = \sum_{j \geq 0} \frac{z^j}{r+j}. \quad (10.1)$$

For each fixed k , counting ordered collections of k blocks and then dividing by $k!$ gives

$$\sum_{N \geq 0} \left\{ \begin{matrix} N \\ k \end{matrix} \right\}_{\geq r} \frac{z^N}{N!} = \frac{1}{k!} \left(\sum_{m \geq r} \frac{z^m}{m!} \right)^k, \quad (10.2)$$

$$\sum_{N \geq 0} \left[\begin{matrix} N \\ k \end{matrix} \right]_{\geq r} \frac{z^N}{N!} = \frac{1}{k!} \left(\sum_{m \geq r} \frac{z^m}{m} \right)^k. \quad (10.3)$$

For cycles, the coefficient $1/m$ is $(m-1)!/m!$. Substituting $N = n + (r-1)k$ and factoring out z^{rk} gives

$$S_{n,k}^{(r)} = \frac{(n + (r-1)k)!}{k!} [z^{n-k}] f_r(z)^k, \quad (10.4)$$

$$C_{n,k}^{(r)} = \frac{(n + (r-1)k)!}{k!} [z^{n-k}] g_r(z)^k. \quad (10.5)$$

No convergence assertion is needed, because each coefficient uses finitely many terms; only degrees through $n - k$ need be retained. These formulas are useful independent checks because they do not use the two-term recurrence, and they reproduce the first-column and diagonal formulas immediately. They test a different thing from the block-size profile formula (2.10): the generating function on one side, the raw combinatorial definition on the other. [D4]

10.2 Row-polynomial operators

Let $\vartheta = x d/dx$, and write $T_n^{(r)}(x) = \sum_k T_{n,k}^{(r)} x^k$ and $C_n^{(r)}(x) = \sum_k C_{n,k}^{(r)} x^k$. Summing the recurrences, with multiplication by x performed *after* the differential operator, gives

$$T_n^{(r)}(x) = xP_d(n+d-1+d\vartheta)T_{n-1}^{(r)}(x) + \vartheta T_{n-1}^{(r)}(x), \quad (10.6)$$

$$C_n^{(r)}(x) = xP_d(n+d-1+d\vartheta)C_{n-1}^{(r)}(x) + (n-1+d\vartheta)C_{n-1}^{(r)}(x). \quad (10.7)$$

Here a polynomial in ϑ acts diagonally on monomials: $P_d(n+d-1+d\vartheta)x^j = P_d(n+d-1+dj)x^j$. Thus the order and indexing of the operator are unambiguous. These formulas offer a compact generating-polynomial description, but the proof above does not require an analysis of polynomial zeros — which, by example 5.2, would in any case be an analysis of nonreal zeros.

10.3 Small rows

Table 2 displays a few of the rows covered by the conjectural cases. Empty places denote entries outside the triangle, not omitted data. The zero column has been suppressed.

r	n	$S_{n,1}^{(r)}$	$S_{n,2}^{(r)}$	$S_{n,3}^{(r)}$	$S_{n,4}^{(r)}$
3	2	1	10		
3	3	1	35	280	
3	4	1	91	2100	15400
4	2	1	35		
4	3	1	126	5775	
4	4	1	336	45045	2627625
5	2	1	126		
5	3	1	462	126126	
5	4	1	1254	1009008	488864376

Table 2: Higher-order subset rows, computed exactly from (2.2) and checked independently using (10.4) and (2.10).

For $r = 3$, the first six subset row sums, starting at $n = 0$, are

$$1, 1, 11, 316, 17592, 1612206.$$

For $r = 4$ they are

$$1, 1, 36, 5902, 2673007, 2582136349,$$

and for $r = 5$,

$$1, 1, 127, 126589, 489874639, 5201522272948.$$

The archive includes row sums through $n = 30$ and complete small triangles through $n = 12$ for both kinds and $1 \leq r \leq 8$. These are reproducible integer-sequence data; no assertion is made about whether each sequence already has a separate OEIS entry.

11 Exact computational audit

The mathematical proof is finite algebra plus induction over all rows. The computations serve two distinct purposes: checking those finite algebraic certificates, and testing for

implementation or indexing mistakes in the number arrays. Neither numerical root finding nor floating-point inequalities are used anywhere. Two independent toolchains are supplied, so that no single implementation — and no single computer algebra version — is the sole witness of any identity. [D5]

11.1 Polynomial certificates without a computer algebra system

The script `certificates.py` implements elementary polynomial addition, multiplication, scalar division, and substitution over $\mathbb{Q}$. Polynomials are tuples of rational coefficients in ascending degree. All identities are checked by exact equality of the coefficient tuples, not by evaluation at finitely many points. The recorded run verified 51 identities or shifted-coefficient positivity certificates, as well as 24 leading-coefficient checks for $d = 1, \dots, 8$.

In particular it verifies (4.13), (4.14), the three coefficientwise identities that together constitute (4.12), (4.16), (7.5), (8.2), and (6.5); it also checks every polynomial listed in section A. The shifted polynomials used for positivity have nonnegative coefficients and positive constant term, so their positivity on a nonnegative half-line is immediate. The full coefficient arrays are written to `data/certificates.json`.

11.2 The same certificates under SymPy

The script `code/verify_certificates.py` performs 24 named exact checks with SymPy, expanding polynomial differences over the rationals and requiring exact zero. Its list includes the universal four-term decomposition (3.4), the lower-order expansions of A_d and H_d , the order-three and order-four boundary bounds, the mixed-term remainder (4.13), the discriminant factorization (4.14), the square certificate (4.12) in one piece, the positive quadratic (4.15) and shifted cubic (4.16), the cycle mixed term and obstruction (8.2), and the sharpness polynomial (6.5). Its report is `data/certificate_checks.json`, which records the Python and SymPy versions actually used. The two checkers are deliberately not a single program: the CAS-free path certifies every identity in this manuscript on its own, and the SymPy path checks the bivariate identity (4.12) directly rather than through its three coefficientwise components.

11.3 Independent integer-array checks

The script `verify.py` streams rows using (2.2) and (2.3). With its default settings, it checks all $n \leq 200$ for $1 \leq r \leq 5$, both subset and cycle triangles. For each of the ten triangles there are

$$\sum_{n=3}^{200} (n-2) = 19701$$

positive-support interior inequalities, giving 197010 exact strict inequalities in total. All passed. The order-five subset lower bound (5.2) was also checked at all 19701 such indices. The order-five cycle run is explicitly marked *experimental* in the output; it is not part of the proved theorem.

The script `code/verify_rows.py` performs a complementary scan that is deeper but narrower: all $n \leq 300$ for the five subset triangles, that is 44850 internal inequalities for each order and 224250 in total. Neither scan range contains the other, and both are run. [D6] That script also evaluates the *full* six-term certificate (4.18), not merely its sign, at all 1711 pairs with $3 \leq n \leq 60$ and $2 \leq k \leq n-1$, checking the identity and each summand's sign separately; it checks the third-row closed form (6.2) for $1 \leq r \leq 100$; and it locates the first failure for

each order $r = 6, \dots, 20$, recording full triples and defects in `data/row_checks.json`. The counterexample table of `data/verification.json` covers $r = 6, \dots, 12$ and is retained beside it rather than replaced. [D10]

As structurally independent entry checks, neither of which uses the recurrence, `verify.py` computes (10.4) and (10.5) by truncated convolution of rational coefficient lists, comparing every entry for $n \leq 12$, $1 \leq r \leq 8$ and both kinds (1456 entries, all agreeing), while `code/verify_rows.py` enumerates unordered block-size profiles and evaluates (2.10) directly (728 entries for $1 \leq r \leq 8$, $0 \leq n \leq 12$, all agreeing). First-column formulas, diagonal formulas, the exact counterexamples, and the operator obstruction of section 8 are also checked. [D4]

Check	Number	Result
CAS-free polynomial identities and positivity certificates	51	Pass
CAS-free leading-coefficient checks, $d = 1, \dots, 8$	24	Pass
SymPy exact identities and sign certificates	24	Pass
Strict interior inequalities, ten triangles, $n \leq 200$	197010	Pass
Quantitative fifth-order bounds, $n \leq 200$	19701	Pass
Strict row inequalities, five subset triangles, $n \leq 300$	224250	Pass
Independent generating-function entries, both kinds	1456	Pass
Independent block-profile entry counts	728	Pass
Full six-term certificate evaluations	1711	Pass
Third-row formula, orders 1 through 100	100	Pass
First-failure tests, orders 6 through 20	15	Pass

Table 3: The verification actually executed for this merged report.

Deb and Sokal already report finite log-concavity testing through $n = 1000$ for the cases $r = 3, 4, 5$ [1, Appendix C, Tables 1–2]. Neither of our smaller independent runs is therefore a claim to improve that computational range. Their purpose is to audit these implementations and the newly derived quantitative bound; the all- n result is supplied by the proof.

11.4 Reproduction and trust boundary

The standard-library scripts require only Python 3.9 or later (`code/verify_rows.py` uses Python 3.10 syntax) and no third-party package:

```
python certificates.py
python verify.py
python code/verify_rows.py --max-n 300 --output-dir data
```

The SymPy checker is the one component with an external dependency; the run recorded here used Python 3.14.4 with SymPy 1.14.0, the version pinned in `requirements.txt` (the second source draft recorded the same SymPy version under Python 3.13.5):

```
python -m pip install -r requirements.txt
python code/verify_certificates.py --output data/certificate_checks.json
```

A longer finite regression run is available explicitly:

```
python verify.py --max-n 500
```

No network request, hidden data file, or proprietary computer algebra service is required, and every identity in this manuscript is verified on the CAS-free path alone. The assertions are implemented as explicit checks and are not disabled by Python’s `-O` flag.

The recurrence computation uses $O(N^2)$ integer-arithmetic operations and stores $O(N)$ entries when rows are streamed, for fixed order r and maximum row N . These are arithmetic-operation counts, not bit-complexity claims: the integers grow with N , and `data/row_checks.json` records the largest entry bit lengths actually exercised (for example 9556 bits at order five, $n = 300$). The independent convolution and block-profile checks are intentionally restricted to small rows, where exact arithmetic is cheap and implementation independence is more valuable than speed.

The scripts are not a proof-assistant formalization. Their trust boundary is ordinary Python integer/rational arithmetic, the small supplied polynomial implementation, and — for the SymPy checker alone — that library’s expansion routines. The paper’s proofs do not logically depend on executing the scripts: all identities, factor signs, support conditions, and induction steps are written out.

12 Conclusion

The chosen problem has a clean resolution: the higher-order Stirling subset triangle has log-concave rows exactly for orders one through five. The proof is short at its core. Normalize the recurrence, keep all terms in the local quadratic bound, use the triangular constraint $t - 5k = n - k - 1 \geq 0$, and apply one identity, (4.12), which converts a negative mixed coefficient from an obstruction into a controlled term of a positive quadratic. A single sixth-order counterexample plus a monotone factorial ratio supplies the sharp converse.

The method also settles the third- and fourth-order cycle cases and yields a continuous interpolating family, and the asymptotics of section 9 explain why the mechanism must change exactly between orders five and six. Its failure at the fifth-order cycle operator is genuine and explicitly certified, not merely a failure to find a convenient factorization. Any extension to that remaining case must exploit a stronger property of the actual preceding rows. The present report separates that open continuation from the completed subset-number proof. Independent mathematical review and a formal proof-assistant development would be appropriate further checks.

A Polynomial identities in one place

This appendix collects the low-degree expressions used in the proof. The independent variable is t , and $H_d(t, k)$ is affine in k . The notation matches the main text; the code uses C0 and C1 for the constant and linear-in- k parts of H_d .

A.1 Pure-square coefficient

$$\begin{aligned} A_0(t) &= 0, \\ A_1(t) &= 1, \\ A_2(t) &= 4(2t^2 - 2t - 3), \\ A_3(t) &= 9(3t^4 - 12t^3 - 9t^2 + 42t + 40), \\ A_4(t) &= 32(2t^6 - 18t^5 + 17t^4 + 168t^3 - 55t^2 - 834t - 630). \end{aligned}$$

A.2 Subset mixed coefficient and its lower bound

$$\begin{aligned}
H_0(t, k) &= 0, \\
H_1(t, k) &= 2, \\
H_2(t, k) &= 4(-2k + 2t - 1), \\
H_3(t, k) &= 6(-9kt + 9k + 3t^2 - 6t + 11), \\
H_4(t, k) &= 16(-12kt^2 + 36kt - 54k + 2t^3 - 9t^2 + 43t - 51).
\end{aligned}$$

The substitution $k = t/(d + 1)$ gives

$$\begin{aligned}
L_0(t) &= 0, & L_1(t) &= 2, \\
L_2(t) &= \frac{4}{3}(4t - 3), \\
L_3(t) &= \frac{3}{2}(3t^2 - 15t + 44), \\
L_4(t) &= -\frac{16}{5}(2t^3 + 9t^2 - 161t + 255) = -\frac{16}{5}\alpha(t).
\end{aligned}$$

A.3 Cycle mixed coefficient

$$\begin{aligned}
M_0(t) &= 0, & M_1(t) &= 2, \\
M_2(t) &= 8(t - 1), \\
M_3(t) &= -18(3t - 11), \\
M_4(t) &= -32(t - 2)(2t^2 + 4t - 51).
\end{aligned}$$

A.4 The three fifth-order auxiliaries

$$\begin{aligned}
\alpha(t) &= 2t^3 + 9t^2 - 161t + 255, \\
\beta(t) &= 2t^2 - 6t + 9 = 2(t - 3/2)^2 + 9/2, \\
\gamma(t) &= 7t^3 - 31t^2 - 126t + 1080, & \gamma(10 + s) &= 7s^3 + 179s^2 + 1354s + 3720.
\end{aligned}$$

Also $\Delta_4(t) = 96\beta(t)$, which is the slope identity behind (4.13).

A.5 A direct expansion check for the main discriminant

For readers checking (4.14) without software, both sides expand to

$$\begin{aligned}
4A_4 - L_4^2 &= \frac{1}{25}(5376t^6 - 66816t^5 + 198528t^4 + 1018368t^3 \\
&\quad - 7986816t^2 + 18351360t - 18662400).
\end{aligned}$$

This displayed coefficient form is itself included in the certificate checks. The factored form is preferable for proving its sign, because it separates the domain condition $t \geq 10$ from the algebraic identity. [D1]

B Deliberate redundancies

Merging the two source packages removed duplicated prose but deliberately kept every place where the two drafts checked the same fact in genuinely different ways. Collapsing any row below would lose an independent check.

Tag	Kept in more than one form	What the duplication buys
D1	The fifth-order certificate as the single identity (4.12), as the two-step discriminant plus completed square (4.14)–(4.17), and as the expanded coefficient form in section A.	The identity is the cleanest to audit, the two-step form is the one that generalizes to the cycle case and to arbitrary d , and the expanded form is the only one checkable entirely by hand.
D2	Two routes to $H_4 \geq L_4$: the exact identity (4.13), and the affine-in- k monotonicity of (4.5) via $\Delta_d > 0$ (lemma 4.1, remark 4.2).	The identity is the shorter proof at $d = 4$; the convexity route is the only one that survives to general d , and theorem 7.1 and section 9 both need it.
D3	Two derivations of $S_{4,2}^{(6)}$ and $S_{4,3}^{(6)}$: block-size pattern and recurrence (section 6).	This is the single load-bearing numeric fact of the converse; a transcription error in either derivation would be caught by the other.
D4	Two entry-verification routes that avoid the recurrence: truncated generating-function convolution (10.4) and block-size profiles (2.10).	One tests the generating function, the other the raw combinatorial definition; an indexing error common to both is very unlikely.
D5	Two verification toolchains: <code>certificates.py</code> (own exact rational polynomial arithmetic, no CAS) and <code>code/verify_certificates.py</code> (SymPy).	The shared identities are confirmed by two independent implementations, and the package is not hostage to one CAS version.
D6	Two finite scan ranges, neither containing the other: $n \leq 200$ across ten subset and cycle triangles, and $n \leq 300$ across the five subset triangles.	The first covers the cycle arrays and the quantitative bound; the second goes deeper in n and exercises much larger integers.
D7	Two witnesses for the fifth-order cycle obstruction: the factorization (8.2) and the exact numerical input at $n = 5$, $q = 7350$.	The algebraic witness shows the criterion fails identically for $t \geq 10$; the numerical one exhibits an actual log-concave input whose image is not log-concave.
D8	Two attributions of the coefficient-sign condition: Sagan [3, Theorem 1] and Shankar [4, Theorem 1.2].	The same sufficient condition is recorded in two independent sources, and each framing of how the present criterion relaxes it is worth keeping.
D9	Two presentations of the sharpness ratio: $F(r)$ decreasing and $R(r) = 1/F(r)$ increasing, with both ratio expressions in (6.4).	Each ratio expression is a transcription check on the other, and both routes reach the same difference polynomial (6.5).
D10	Two recorded counterexample tables: <code>data/verification.json</code> for $r = 6, \dots, 12$ and <code>data/row_checks.json</code> for $r = 6, \dots, 20$.	The union is kept; the second extends the range and adds bit-length instrumentation, the first adds the cycle-operator obstruction record.
D11	Two shape consequences: ratio monotonicity with the q -weighted log-concave mass function, and the balanced-product inequalities (5.4).	Neither statement implies the other as stated: the balanced products are stronger as inequalities, the weighted-distribution remark is about a different object.

Tag	Kept in more than one form	What the duplication buys
D12	Two scope fences: the result-to-source map in section D and the nonreal-rootedness example 5.2.	The first guards against overclaiming which conjecture clauses are settled; the second rules out the real-rootedness and total-positivity readings of the result.

C A minimal symbolic check

The following compact calculation verifies the central equality (4.12) alone, in the letters of this manuscript. The two full checkers in the archive verify all the auxiliary identities as well, and `certificates.py` does so without SymPy.

```
import sympy as s

t, k, u, v = s.symbols("t k u v")
P4 = t*(t-1)*(t-2)*(t-3)
pm, pp = P4.subs(t, t-4), P4.subs(t, t+4)
A4 = P4*P4 - pm*pp
H4 = 2*k*P4 - (k+1)*pm - (k-1)*pp
alpha = 2*t**3 + 9*t*t - 161*t + 255
beta = 2*t*t - 6*t + 9
gamma = 7*t**3 - 31*t*t - 126*t + 1080
lhs = 25*(A4*u*u + H4*u*v + v*v)
rhs = ((5*v - 8*alpha*u)**2 + 96*(t-5)*beta*gamma*u*u
        + 480*beta*(t - 5*k)*u*v)
assert s.expand(lhs - rhs) == 0
```

An exact polynomial identity is only one part of the proof. Its application also needs the combinatorial recurrence, the support and boundary analysis, the preceding row's log-concavity, and the domain $t \geq 5k \geq 10$. Those mathematical hypotheses are proved explicitly in the main text rather than delegated to this code.

D Source and status notes

The problem identification uses the specific version `arXiv:2507.18959v1`. The conjecture and recurrence were checked in both its HTML and PDF renderings. On the report date the arXiv abstract page showed only the July 2025 version in its submission history. Searches for the exact paper identifier, the phrase “higher-order Stirling” with log-concavity, and relevant arXiv author/topic combinations did not locate a subsequent resolution. The BCC30 statement and the recent triangular-recurrence paper [4] were checked separately. Both source drafts performed such a check independently, on 19 and on 20 September 2026; neither located a later resolution. This is a documented bounded search, not proof of absence from all literature, and it does not establish priority.

The underlying combinatorial objects, their recurrences, and the target conjecture are from the cited literature. What the present report offers as its own contribution is the fifth-order certificate and its two forms, the induction, the quantitative bound, the sharpness argument, the cycle extensions and obstruction, the interpolating family, the asymptotic explanation, and the accompanying checks. No novelty is claimed for the general two-variable preservation lemma, for the elementary identities, or for the small counterexamples.

The mathematical claims established here can therefore be assessed independently of any priority judgment. The result-to-source map is:

Source assertion	Result in this manuscript
Deb–Sokal 1.4(c), row log-concavity	All stated orders $1 \leq r \leq 5$; strictness and sharp converse also proved.
Deb–Sokal 1.3(c), row log-concavity	Orders $1 \leq r \leq 4$ proved; order five not settled.
BCC30 Problem 30.9	Cycle orders three and four proved; order five not settled.
Other clauses of those conjectures	Not claimed as consequences of the results here.

References

- [1] Bishal Deb and Alan D. Sokal, *Higher-order Stirling cycle and subset triangles: Total positivity, continued fractions and real-rootedness*, arXiv:2507.18959v1, 2025. See Lemma 1.1, Conjectures 1.3(c) and 1.4(c), and Appendix C.
- [2] Peter Cameron (editor), *Problems from BCC30*, arXiv:2409.07216v1, 2024. Problem 30.9, presented by Bishal Deb, concerns the cycle triangle and is not a separate source for the subset conjecture.
- [3] Bruce E. Sagan, *Inductive and injective proofs of log concavity results*, Discrete Mathematics **68** (1988), 281–292. doi:10.1016/0012-365X(88)90120-3. See Theorem 1 and its inductive proof.
- [4] Umesh Shankar, *Log-concavity of rows of triangular arrays satisfying a certain super-recurrence*, arXiv:2508.12467v1, 2025. See Theorem 1.2 for the coefficient-sign criterion, Theorems 1.3–1.4 for its relatives, and Section 1 for the different r -Stirling family.
- [5] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A134991: Triangle of Ward numbers, accessed 19 September 2026.